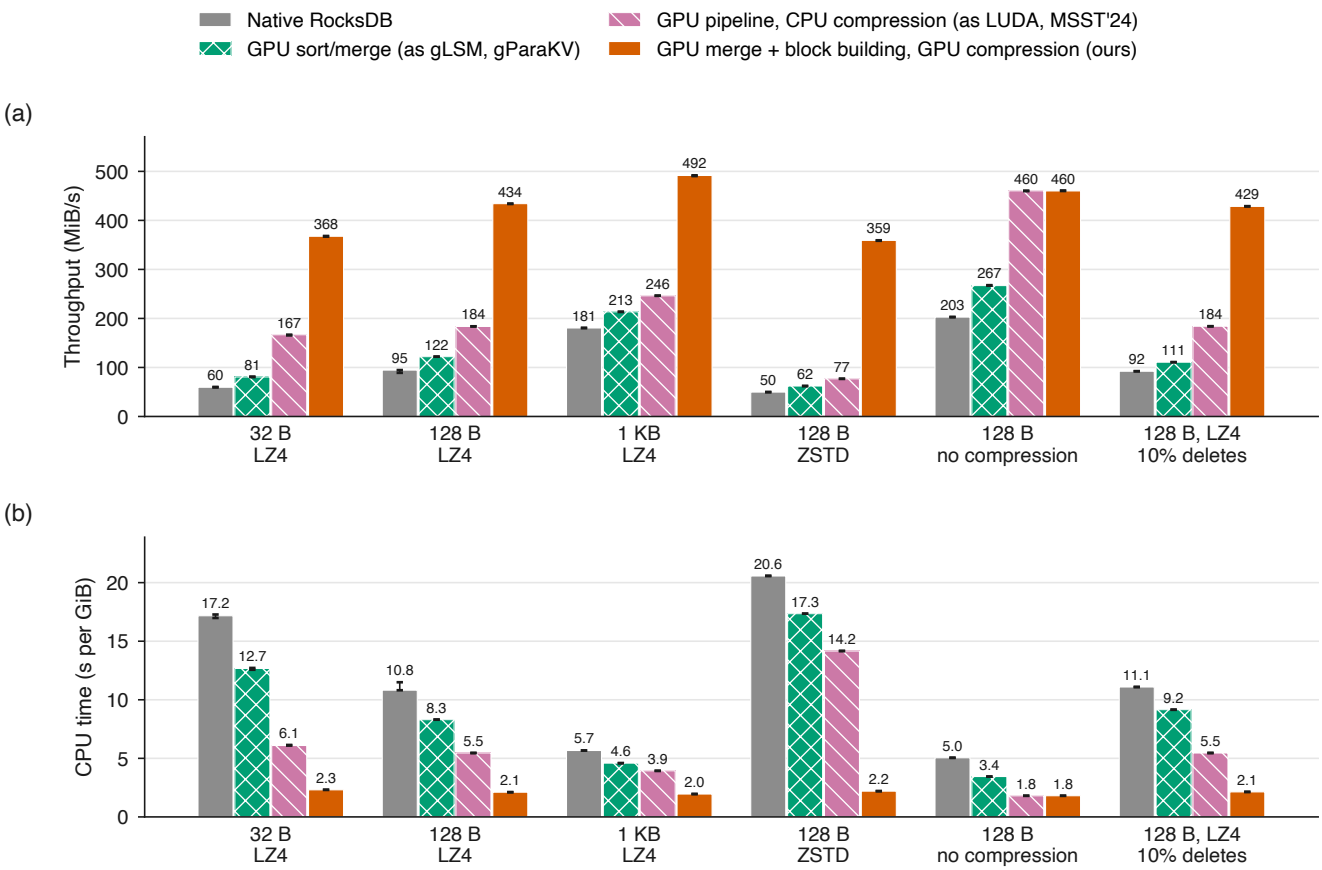
Native RocksDB
GPU sort/merge (as gLSM, gParaKV)
GPU pipeline, CPU compression (as LUDA, MSST'24)
GPU merge + block building, GPU compression (ours)
(a)
Throughput (MiB/s)
0
100
200
300
400
500
60
81
167
368
95
122
184
434
181
213
246
492
50
62
77
359
203
267
460
460
92
111
184
429
32 B LZ4
128 B LZ4
1 KB LZ4
128 B ZSTD
128 B no compression
128 B, LZ4 10% deletes
(b)
CPU time (s per GiB)
0
5
10
15
20
17.2
12.7
6.1
2.3
10.8
8.3
5.5
2.1
5.7
4.6
3.9
2.0
20.6
17.3
14.2
2.2
5.0
3.4
1.8
1.8
11.1
9.2
5.5
2.1
32 B LZ4
128 B LZ4
1 KB LZ4
128 B ZSTD
128 B no compression
128 B, LZ4 10% deletes